DATA PREPROCESSING REPORT

Heart Disease Prediction: Feature Engineering & Preprocessing

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Overview

This report documents every preprocessing decision applied to the Heart Failure Prediction dataset (Kaggle) in preparation for machine learning. The raw dataset contains 918 patient records and 12 clinical and demographic features, with HeartDisease as the binary target variable (1 = heart disease present, 0 = absent). Each stage below explains what was done, why the method was chosen, and what the result was.

Stage	Result
Raw dataset	918 rows x 12 columns
Cleaned dataset	918 rows x 12 columns, 0 missing values
ML-ready	918 rows x 16 columns, fully numeric
Rows removed	0 (no duplicates found; outliers capped, no deleted).

Data Cleaning

Missing Value Assessment

An initial null check **(df.isnull().sum())** returned zero missing values across all columns. However, inspecting descriptive statistics revealed that RestingBloodPressure and Cholesterol contained values of 0, which is not physiologically possible for a living patient. These zeros were identified as disguised missing values rather than genuine measurements.

- RestingBloodPressure: 1 record with a value of 0.
- Cholesterol: 172 records with a value of 0 (approximately 19% of the dataset).

These zeros were converted to NaN, then imputed using the median of each respective column. The median was chosen over the mean because it is more resistant to outliers and skew, which is appropriate for clinical measurements that can vary widely across a patient population.

Duplicate Records

A duplicate check (df.duplicated().sum()) confirmed zero duplicate rows. No records were removed at this stage.

Data Tye Validation

All columns were reviewed for correct data types following cleaning. RestingBloodPressure and Cholesterol were converted to float64 as a result of median imputation; all other columns retained appropriate types (int64 for whole-number measurements, object for categorical text fields prior to encoding).

Feature Engineering

Column Renaming

Four columns were renamed for clarity and readability: FastingBS to FastingBloodSugar, RestingBP to RestingBloodPressure, RestingECG to RestingECGResult, and MaxHR to MaxHeartRate.

New Feature Created

An AgeGroup feature was engineered by binning Age into four categories (Under 40, 40-50, 50-60, 60+), to test whether age bracketing revealed patterns not visible in the continuous Age variable. This feature was later assessed during Feature Selection and ultimately removed due to redundancy with Age.

Irrelevant Feature Checking

Each column was checked for near-zero variance (a single dominant value across nearly all records), which would indicate no predictive value, similar to constant columns identified in prior work. No such columns were found; every original feature in this dataset carries clinically meaningful variation and was retained.

Feature Encoding

Two encoding techniques were applied, matched to the structure of each categorical variable.

Label Encoding

Applied to Sex and ExerciseAngina, both binary variables (two categories only). Label Encoding was appropriate here because with only two possible values, there is no risk of implying a false order between categories, the encoding is simply a substitution (e.g., M = 1, F = 0).

One-Hot Encoding

Applied to ChestPainType, RestingECGResult, ST_Slope, and AgeGroup, all variables with three or more unrelated categories. Label Encoding these would have introduced a false numerical order (for example, implying ChestPainType category 2 is somehow "greater" than category 1), which does not reflect reality. One-Hot Encoding avoids this by creating independent binary columns per category. drop_first=True was used throughout to avoid the dummy variable trap, where the dropped category becomes the implied reference group.

Feature Scaling

Five continuous numeric columns (Age, RestingBloodPressure, Cholesterol, MaxHeartRate, Oldpeak) were scaled using StandardScaler, which transforms each column to a mean of 0 and standard deviation of 1. This method was selected because these features sit on very different original scales (for example, Cholesterol ranges into the hundreds while Oldpeak ranges roughly from -2.6 to 6.2), and many machine learning algorithms are sensitive to this kind of scale imbalance. After scaling, all five columns showed a mean of approximately 0 and a standard deviation of approximately 1, confirming the transformation was applied correctly.

Outlier Detection and Treatment

Outliers were assessed using independent methods for cross-validation.

IQR Method

The interquartile Range method (values beyond Q1 -1.5x IQR or Q3 +1.5x IQR) was applied to the five continuous columns:

Column	Outliers Detected
Age	0
RestingBloodPPressure	41
Cholesterol	41
MaxHeartRate	2
Oldpeak	16

Treatment Approach

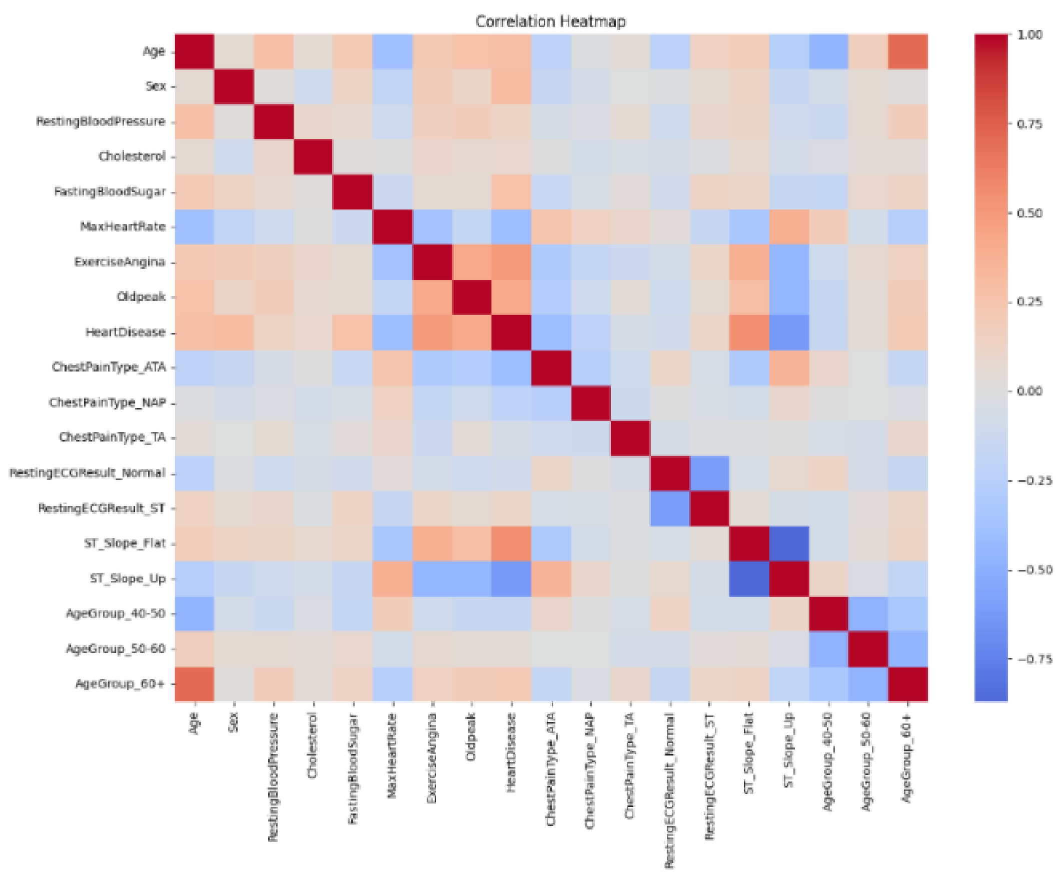
Rather than removing outlier rows, values were capped (Winsorized) to the IQR boundaries. This decision was made because extreme readings in blood pressure, cholesterol, and Oldpeak are clinically plausible in high-risk cardiac patients; removing these rows risked discarding the exact patient profiles the eventual model needs to learn to identify. Capping reduces the statistical influence of extreme values while preserving all 918 patient records.

Z-Score

As a second, independent check, Z-scores were calculated for the same five columns, flagging any value with an absolute Z-score above 3. After IQR capping, only one residual outlier remained (in Oldpeak), confirming the capping approach was effective.

Feature Selection

Correlation with Target Variable



A correlation heatmap is used to identify the most important features. The strongest predictors identified were:

- ST_Slope_Flat (+0.55) and ST_Slope_Up (-0.62)
- ExerciseAngina (+0.49)
- Oldpeak (+0.41)
- MaxHeartRate (-0.40) and ChestPainType_ATA (-0.40)
- Sex (+0.30) and Age (+0.28)

Cholesterol, RestingECGResult categories, and some AgeGroup bins showed weak correlation with the target (below 0.10) but were not removed on this basis alone, since weak individual correlation does not rule out predictive value in combination with other features.

Redundant Feature Check

A pairwise correlation check (threshold above 0.7) identified two highly correlated pairs:

- ST_Slope_Up and ST_Slope_Flat (0.87): expected, since both originate from one-hot encoding the same original ST_Slope column. Retained, as this is a normal encoding artifact rather than true redundancy.
- Age and AgeGroup_60+ (0.70): genuine redundancy, since AgeGroup was engineered directly from Age.

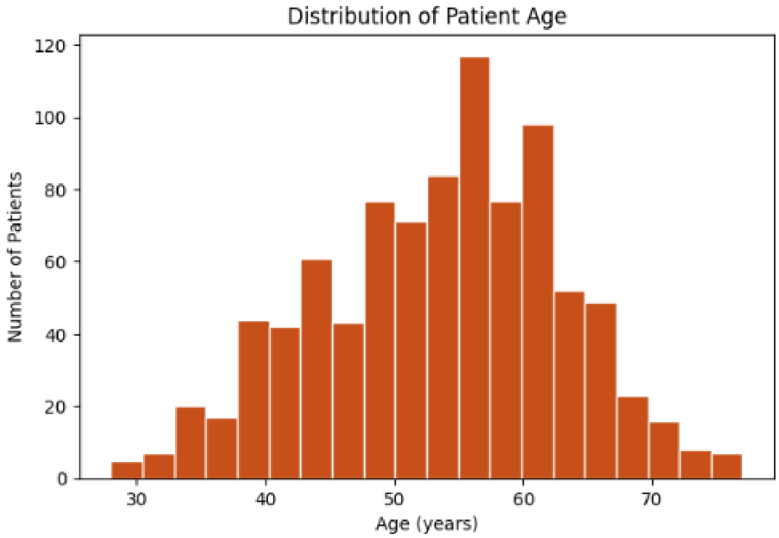
Feature Removed

The three AgeGroup dummy columns (AgeGroup_40-50, AgeGroup_50-60, AgeGroup_60+) were dropped, retaining the more precise continuous Age column instead. This reduced the dataset from 19 to 16 columns without losing any specific or unique information.

Visualizations

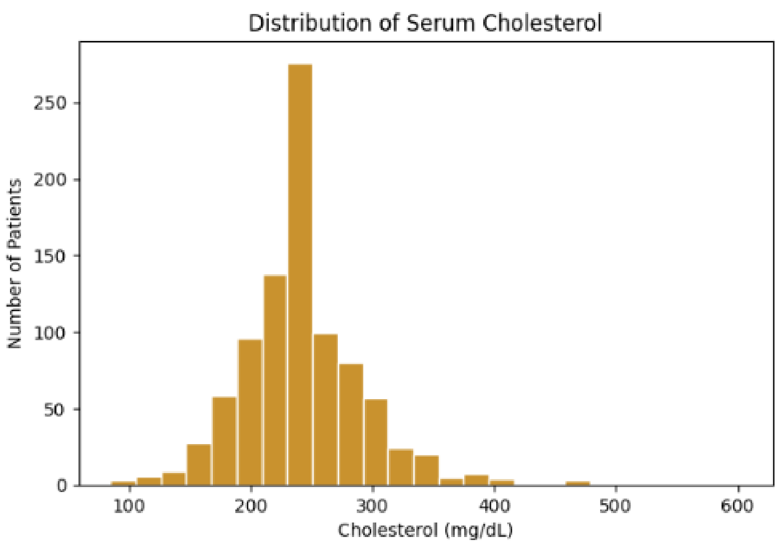
The following visualizations were produced to explore feature distributions and their relationship with the target variable.

Histogram: Distribution of Patient Age



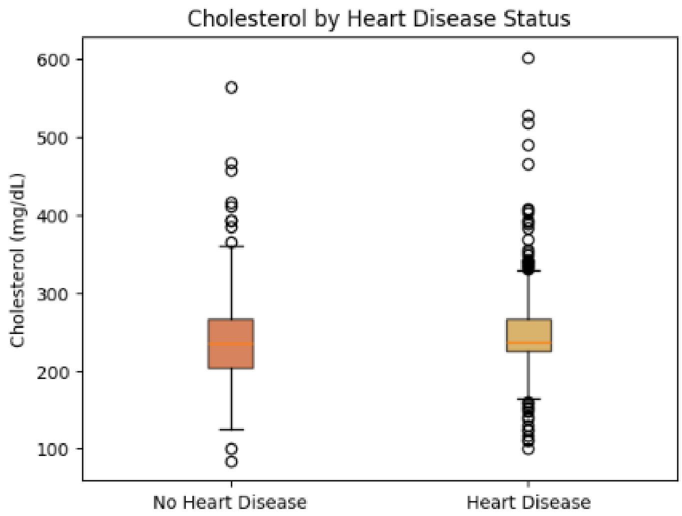
The patient population is concentrated between roughly ages 45 and 60, with a fairly normal, slightly left-skewed distribution.

Histogram: Distribution of Serum Cholesterol



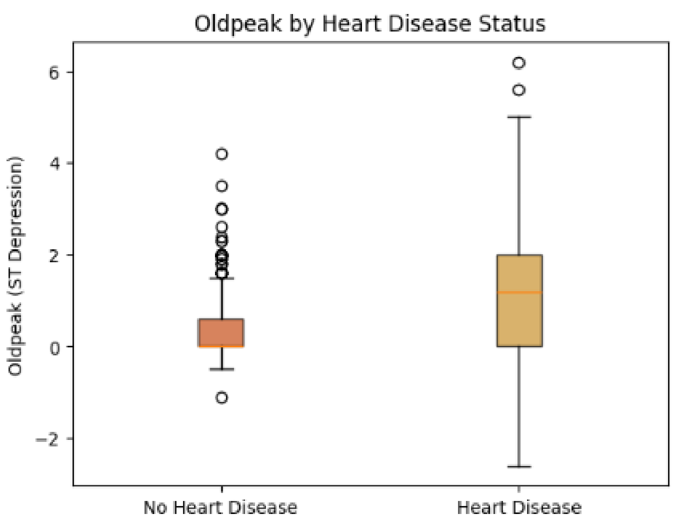
A visible spike sits at the median-imputed value, reflecting the 172 records that were filled during cleaning. The remaining distribution is right-skewed with a long tail toward higher readings.

Boxplot: Cholesterol by Heart Disease Status



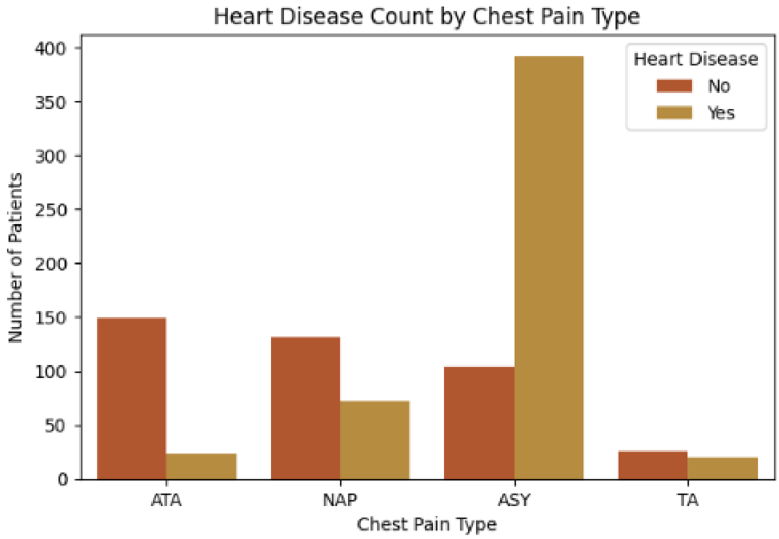
Median cholesterol is similar between groups (235 vs 237 mg/dL), consistent with the weak correlation.

Boxplot: Oldpeak by Heart Disease Status



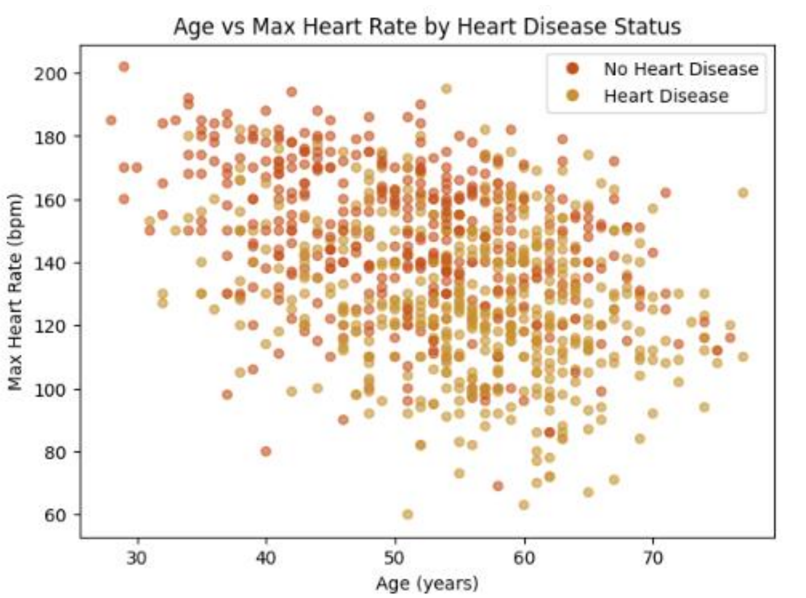
Patients with heart disease show a clearly higher median Oldpeak (1.2) than those without (0.0), consistent with Oldpeak being one of the strongest predictors identified.

Count Plot: Heart Disease Count by Chest Pain Type



Asymptomatic chest pain (ASY) shows a disproportionately high count of heart disease cases, a clinically notable finding since the absence of typical chest pain symptoms does not indicate absence of disease.

Scatter Plot: Age vs Max Heart Rate



A negative relationship is visible between age and max heart rate (correlation of -0.38), and heart disease cases (gold) cluster more toward lower max heart rates across most ages.